

Network Latency in Distributed LLM Inference

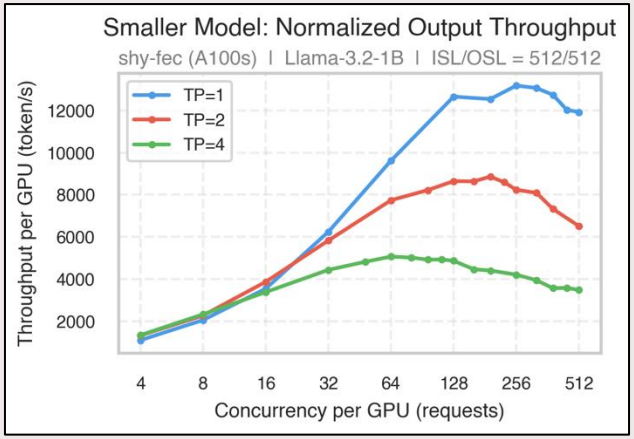
PROGRESS UPDATE: MAY 27, 2026

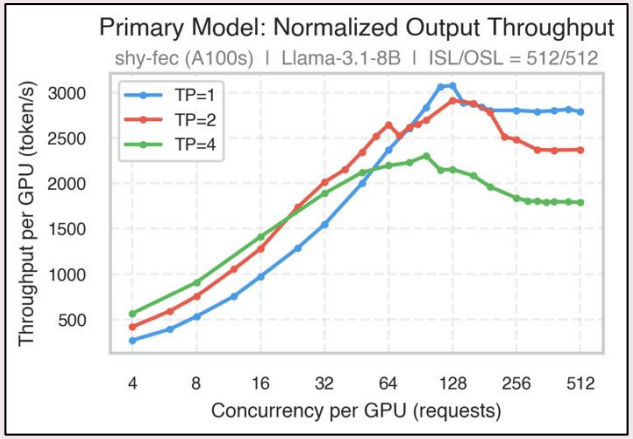
Kelsey Penners

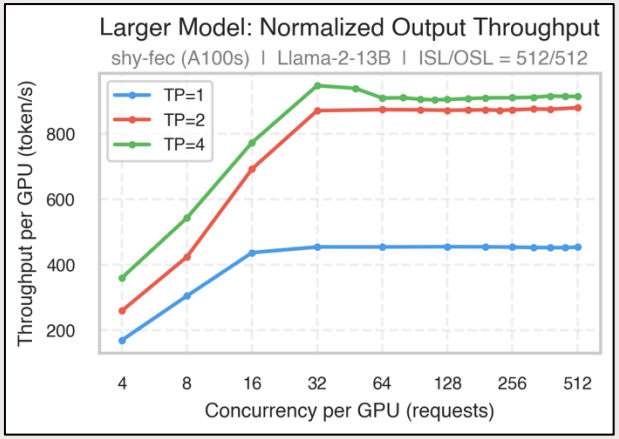
Status / Progress

- TP scaling with 1B and 13B models on both servers
- Interconnect comparison with TP=4
- Hardware measurements: `p2pBandwidthLatencyTest` and `nccl-tests`
- Writing progress:
  - Multi-GPU chapter structured and in progress!
  - Methodology section mostly polished up.
  - Results section will include scaling with tensor parallelism for primary model, then exploring model size sensitivity, followed by interconnect / transport sensitivity and potential SLO analysis subsection.

TP Scaling with Model Sizes: shy-fec



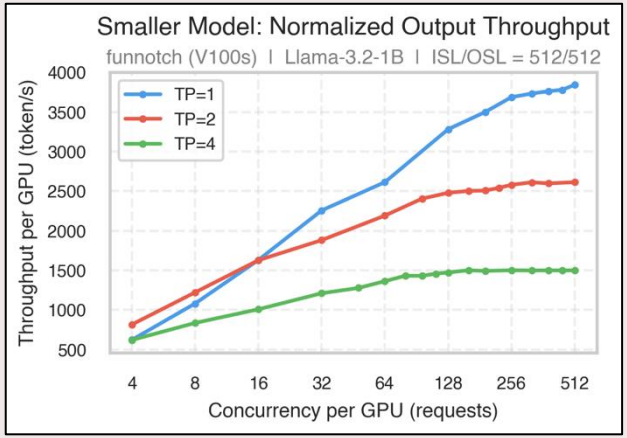


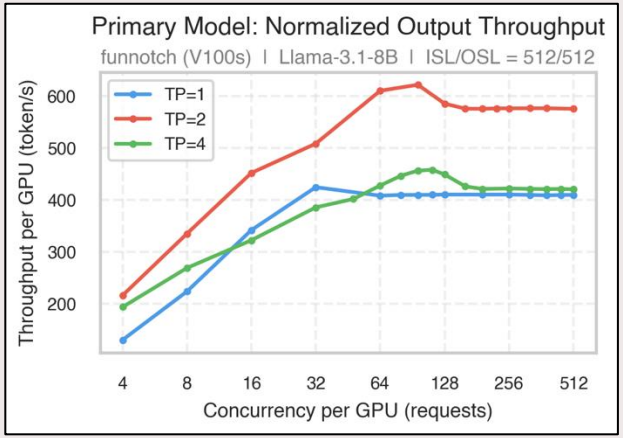


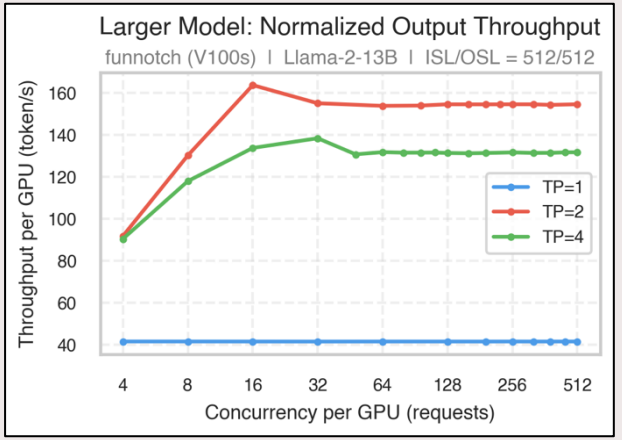
- Small models don't benefit from TP due to communication overhead
- With NVLink, TP=4 in optimal for 13B -> memory and compute benefits outweigh communication cost
- How well an LLM scales relies directly on the size of the model and available VRAM

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TP Scaling with Model Sizes: funnotch

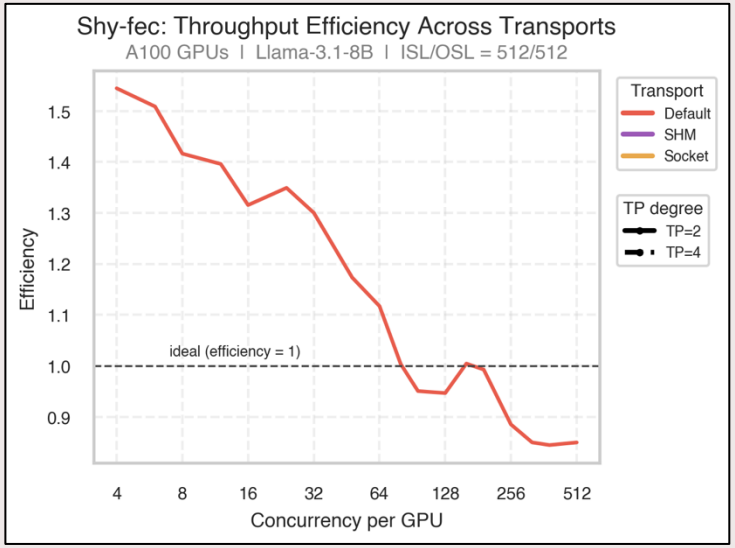


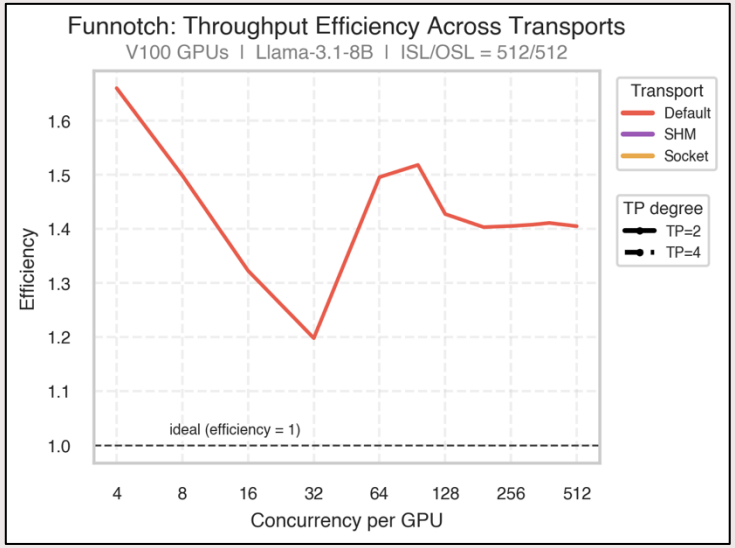




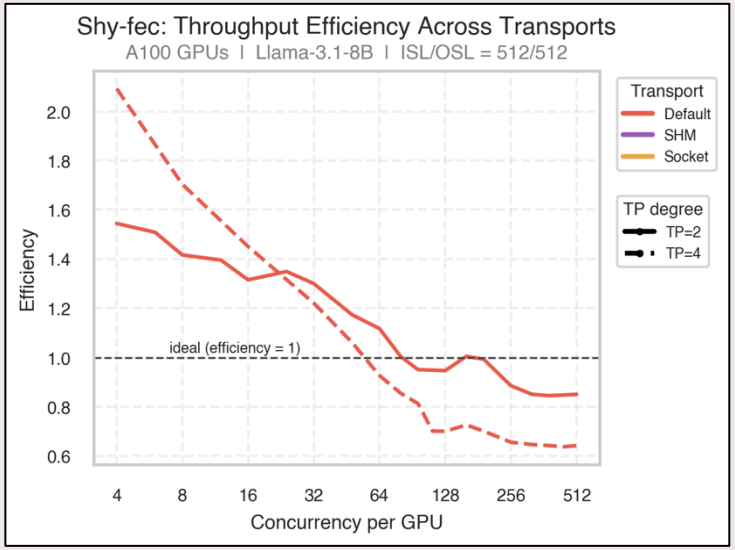
- Again, small model does best on a single GPU
- TP provides benefits for larger models, up to an optimal GPU count
- On PCIe, TP=2 is optimal for 13B -> communication overhead grows faster than on `shy-fec`
  - Communication cost outweighs the extra memory benefit

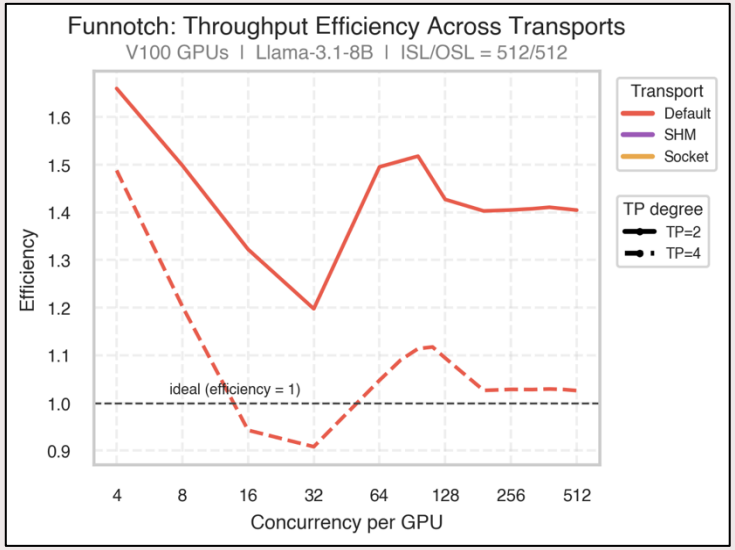
Efficiency over Transports



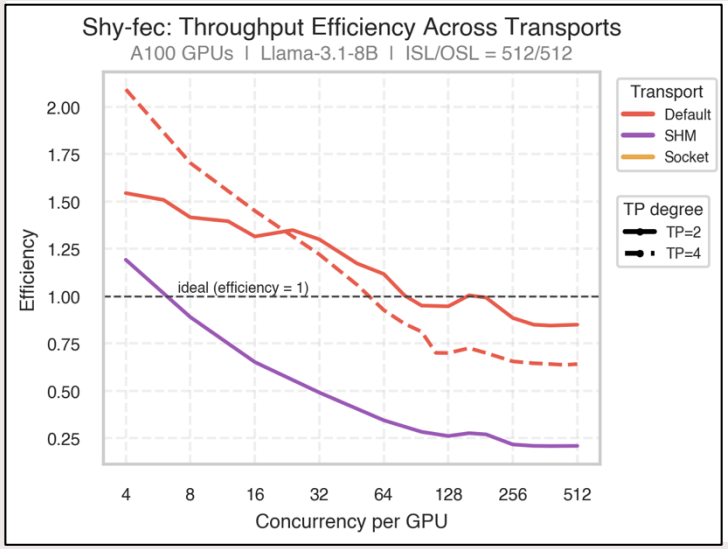


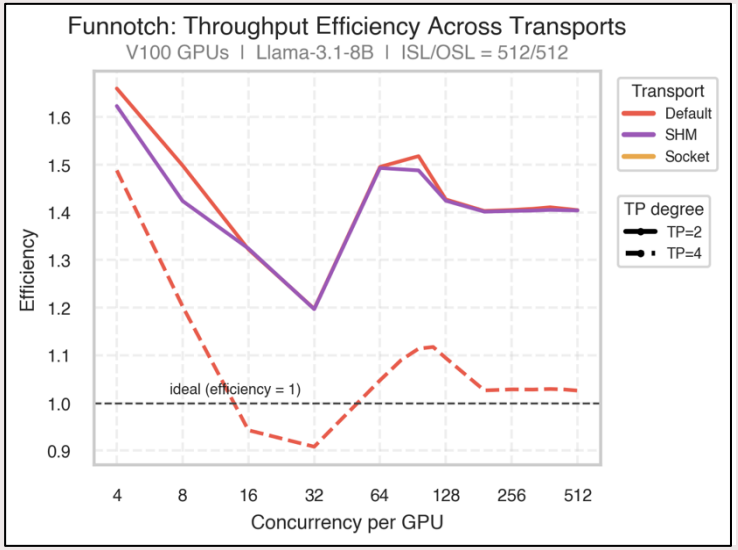
Efficiency over Transports



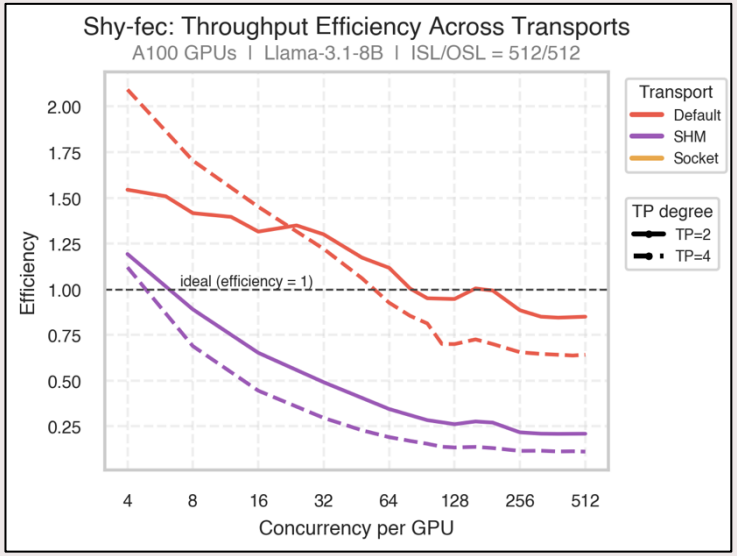


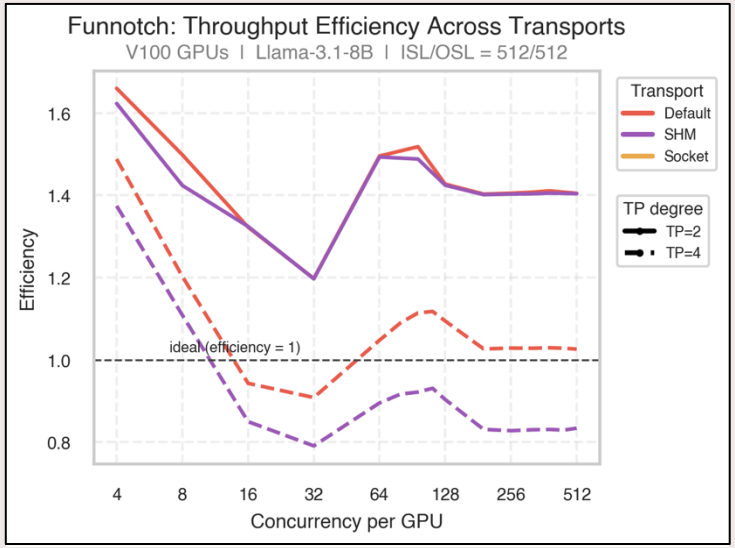
Efficiency over Transports



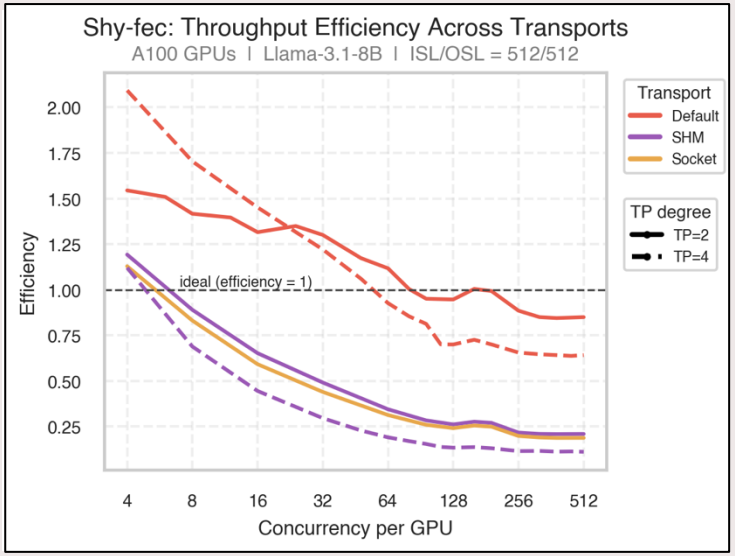


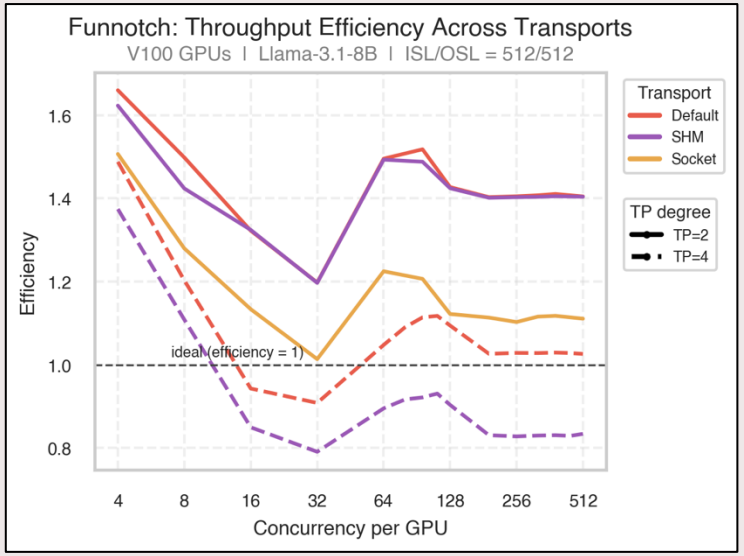
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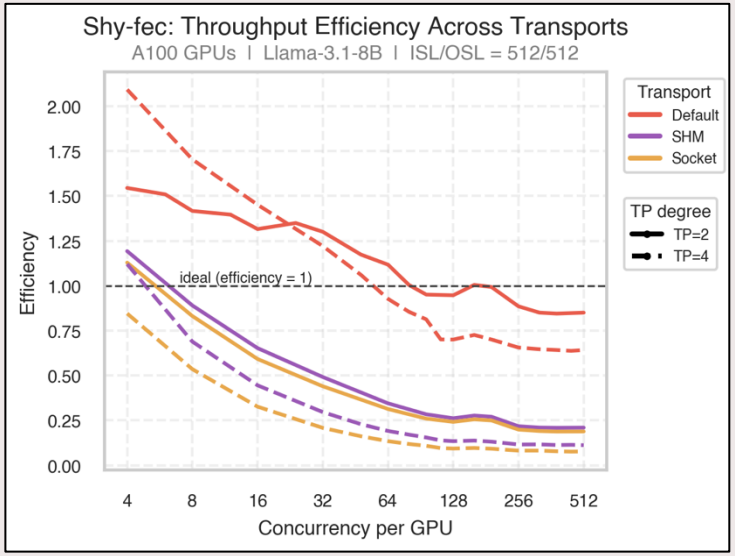


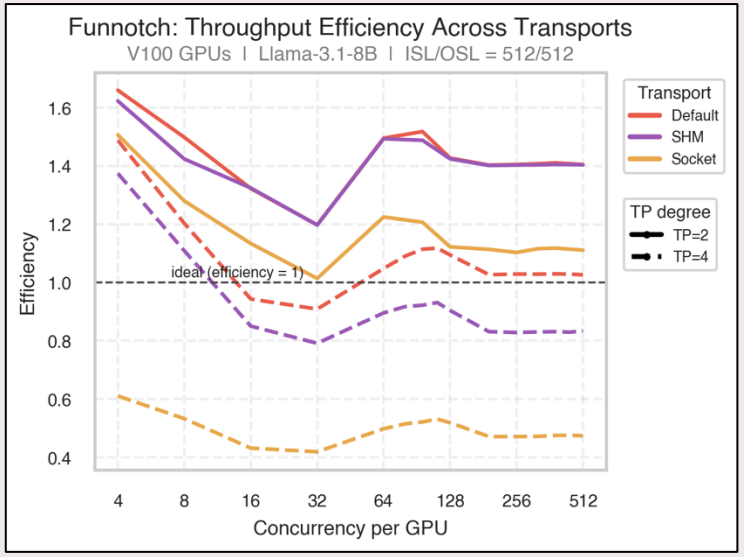
Efficiency over Transports



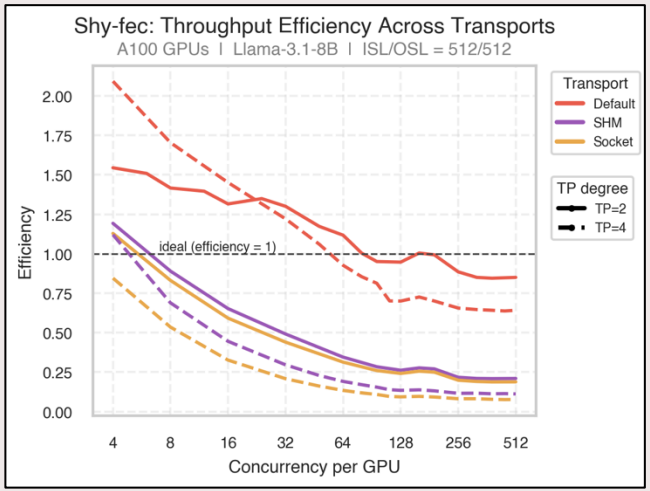


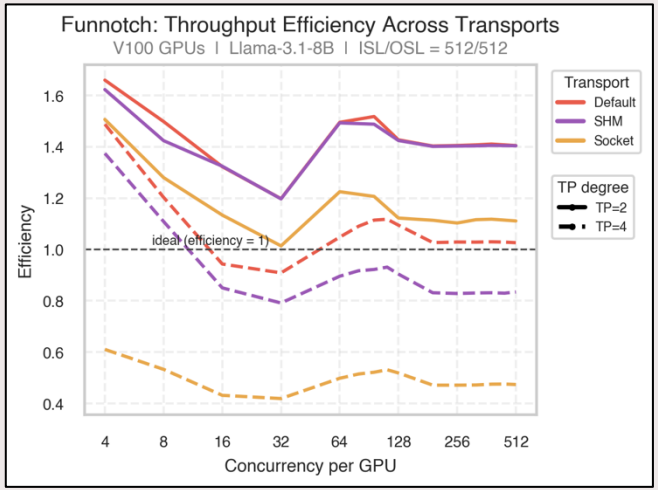
Efficiency over Transports





Efficiency over Transports





- On `shy-fec`, NVLink is the most efficient transport, SHM ≈ Socket across all concurrencies -> PCIe is the bottleneck here
- On `funnotch`, PCIe is the most efficient transport.
- As we increase TP degree, communication overhead compounds

Hardware Measurement Tools

- `p2pBandwidthLatencyTest`:
  - measures GPU-to-GPU data transfer latency and bandwidth
  - direct look at hardware layer
  - lowest boundary for communication cost
- `nccl-tests`:
  - measures collective commuication (e.g., AllReduce op latency)
  - reveals communication cost under tensor parallelism
  - can measure operation latency across various transport configs

P2P=Disabled Latency Matrix (us)

GPU	0	1	2	3	4	5	6	7
0	3.31	25.54	24.79	24.59	25.48	25.54	24.67	26.55
1	25.60	2.95	24.60	24.61	25.48	25.49	24.67	24.68
2	25.52	24.59	3.28	24.60	25.51	25.46	24.62	24.59
3	24.61	24.59	24.66	3.29	25.47	25.18	24.60	24.60
4	24.68	24.61	24.71	24.68	3.13	25.55	26.57	24.70
5	24.70	24.72	24.78	24.69	25.59	3.11	24.92	24.72
6	24.69	24.62	24.60	24.60	25.54	25.35	2.65	24.69
7	24.74	24.59	24.68	24.59	25.20	25.46	24.66	2.42

P2P=Enabled Latency (P2P Writes) Matrix (us)

GPU	0	1	2	3	4	5	6	7
0	3.32	3.46	3.49	3.56	3.56	3.53	3.53	3.46
1	3.56	2.94	3.53	3.50	3.50	3.49	3.55	3.50
2	3.53	3.57	3.26	3.52	3.45	3.51	3.46	3.46
3	3.57	3.53	3.46	3.30	3.54	3.52	3.50	3.50
4	3.58	3.64	3.61	3.63	3.14	3.62	3.67	3.60
5	3.65	3.64	3.63	3.56	3.61	3.11	3.64	3.64
6	3.03	3.03	3.10	3.01	3.08	3.06	2.65	3.09
7	3.06	3.10	3.03	3.02	3.10	3.14	3.09	2.44

```
# Using devices
#  Rank  0 Group  0 Pid 459304 on    shy-fec device  0 [0000:c7:00] NVIDIA A100-SXM4-40GB
#  Rank  1 Group  0 Pid 459304 on    shy-fec device  1 [0000:ca:00] NVIDIA A100-SXM4-40GB
#
#                                                     out-of-place                       in-place
#       size         count      type   redop    root     time   algbw   busbw  #wrong     time   algbw   busbw  #wrong
#        (B)    (elements)                               (us)  (GB/s)  (GB/s)             (us)  (GB/s)  (GB/s)
        4096          1024     float     sum      -1    18.38    0.22    0.22       0    17.82    0.23    0.23       0
        8192          2048     float     sum      -1    21.01    0.39    0.39       0    20.62    0.40    0.40       0
       16384          4096     float     sum      -1    25.96    0.63    0.63       0    26.18    0.63    0.63       0
       32768          8192     float     sum      -1    39.49    0.83    0.83       0    38.79    0.84    0.84       0
       65536         16384     float     sum      -1    68.05    0.96    0.96       0    68.16    0.96    0.96       0
      131072         32768     float     sum      -1   119.82    1.09    1.09       0   120.74    1.09    1.09       0
      262144         65536     float     sum      -1   158.82    1.65    1.65       0   164.80    1.59    1.59       0
      524288        131072     float     sum      -1   300.98    1.74    1.74       0   297.96    1.76    1.76       0
     1048576        262144     float     sum      -1   563.32    1.86    1.86       0   562.61    1.86    1.86       0
     2097152        524288     float     sum      -1  1129.30    1.86    1.86       0  1163.87    1.80    1.80       0
     4194304       1048576     float     sum      -1  2354.72    1.78    1.78       0  2363.92    1.77    1.77       0
     8388608       2097152     float     sum      -1  5076.34    1.65    1.65       0  4999.46    1.68    1.68       0
    16777216       4194304     float     sum      -1  10539.7    1.59    1.59       0  10545.3    1.59    1.59       0
# Out of bounds values : 0 OK
# Avg bus bandwidth    : 1.24889
#
# Collective test concluded: all_reduce_perf
#
```


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Discussion

- Github email?
- What results worthy enough to include in Multi-GPU chapter vs. should be put in appendix?
- Timeline check: currently in week 19

Next Steps

- Finish writing multi-GPU results
- Analytical comparison: translating the transport/interconnect latency measurements into predicted overhead, then compare to these empirical results.

Thank you!